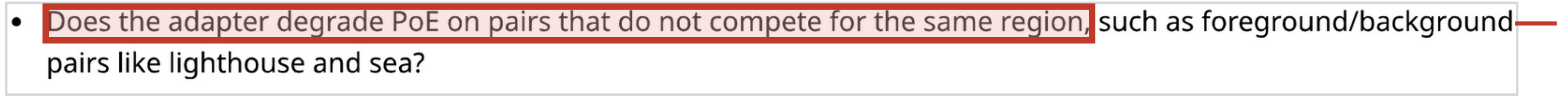
G12: Pairs that do not compete for one region

UHz6 question 4 · status: answered (the reply still has a TODO for CO3's row, now in Table 3)

THE REVIEWER'S POINT

UHz6, question 4



- Does the adapter degrade PoE on pairs that do not compete for the same region, such as foreground/background pairs like lighthouse and sea?

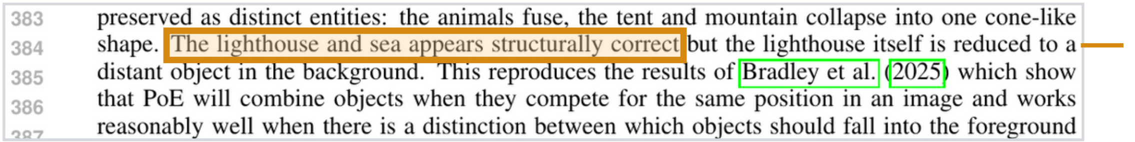
Reviews.pdf, page 3 (no text layer; passage located by OCR)

U-Q4

one image

THE SUBMITTED WEAKNESS

Section 4.1, one lighthouse



preserved as distinct entities: the animals fuse, the tent and mountain collapse into one cone-like shape. The lighthouse and sea appears structurally correct but the lighthouse itself is reduced to a distant object in the background. This reproduces the results of Bradley et al. (2025) which show that PoE will combine objects when they compete for the same position in an image and works reasonably well when there is a distinction between which objects should fall into the foreground

56173_Restoring_plurality_in_c.pdf (submitted), page 8

one example, no count

scored

grouped

THE REVISED ANSWER

Table 3, the not-competing group

21 not-competing pairs

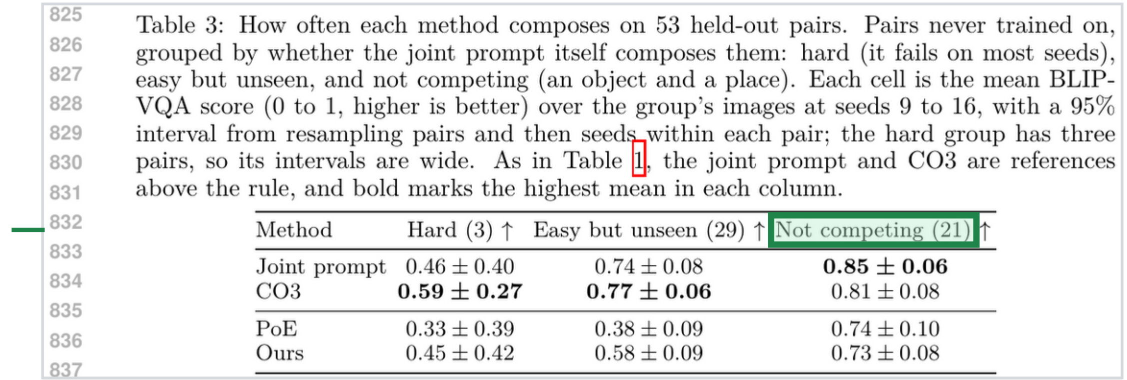


Table 3: How often each method composes on 53 held-out pairs. Pairs never trained on, grouped by whether the joint prompt itself composes them: hard (it fails on most seeds), easy but unseen, and not competing (an object and a place). Each cell is the mean BLIP-VQA score (0 to 1, higher is better) over the group's images at seeds 9 to 16, with a 95% interval from resampling pairs and then seeds within each pair; the hard group has three pairs, so its intervals are wide. As in Table 1, the joint prompt and CO3 are references above the rule, and bold marks the highest mean in each column.

Method	Hard (3) ↑	Easy but unseen (29) ↑	Not competing (21) ↑
Joint prompt	0.46 ± 0.40	0.74 ± 0.08	**0.85 ± 0.06**
CO3	**0.59 ± 0.27**	**0.77 ± 0.06**	0.81 ± 0.08
PoE	0.33 ± 0.39	0.38 ± 0.09	0.74 ± 0.10
Ours	0.45 ± 0.42	0.58 ± 0.09	0.73 ± 0.08

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 16

Appendix C.2, the held-out pairs

the groups

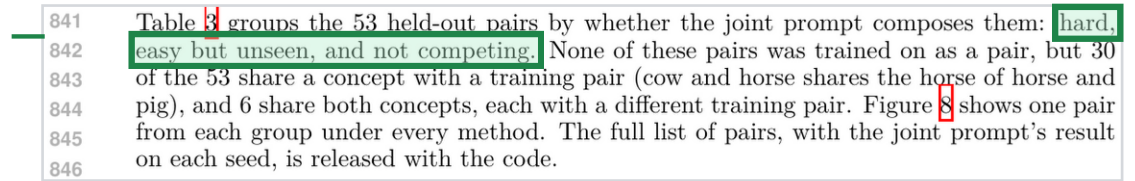


Table 3 groups the 53 held-out pairs by whether the joint prompt composes them: hard, easy but unseen, and not competing. None of these pairs was trained on as a pair, but 30 of the 53 share a concept with a training pair (cow and horse shares the horse of horse and pig), and 6 share both concepts, each with a different training pair. Figure 8 shows one pair from each group under every method. The full list of pairs, with the joint prompt's result on each seed, is released with the code.

paper/iclr-56173/build/iclr2027_conference.pdf (revised), page 16

The reply we plan to send (replies/k4nd-and-uhz6.md: UHz6 Q4)

answered, reply TODO

Q4: pairs that do not compete for the same region

No. Table 3's not-competing group holds 21 held-out pairs whose concepts do not contend for one region, including lighthouse and a stormy sea (Appendix C.2). On these, the adapter scores 0.73 ± 0.08 against the product's 0.74 ± 0.10 (BLIP-VQA, seeds 9 to 16, 95% intervals from resampling pairs). The intervals overlap, so on pairs the product already composes, the adapter neither helps nor harms on this measure. [TODO: CO3's row, slot R3.]

Table 3 now carries CO3's row; the reply's "[TODO: CO3's row, slot R3]" has not been filled yet. Every excerpt is a crop of a page render; the file and page are printed under it. Boxes mark text already on the page.